The Million Image Mountain

Technical challenges and improvements within our bulk digitization projects

UNB LIBRARIES Fredericton

APLA Conference, 2018

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James MacKenzie
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Web Developer

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Manager

Jacob Sanford
Sr. Tech. Operations Manager

Jeremy McDermott
Web Developer

Senior Technical Operations Manager?

https://commons.wikimedia.org/wiki/File:The_Harriet_Irving_Library.JPG

Digitization Projects

Connell Memorial Herbarium

- 40,000 Specimens
- Source : Large Format Cards

New Brunswick Historical Newspapers

- 150,000 Pages
- Source : Physical Paper

The Daily Mail

ROUMANIANS WON A VICTORY OVER GERMANS AND BULGARS

French Troops Hold Ground On Somme Against Huns

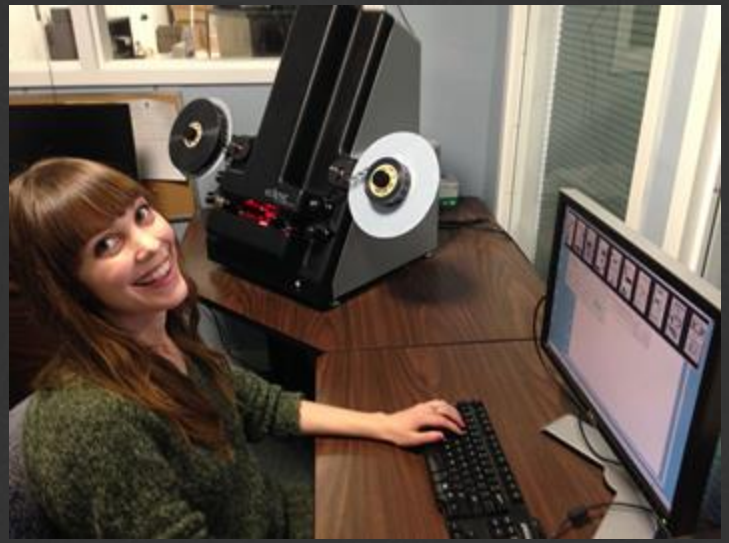
ROUMANIAN FORCES HAVE REPULSED AN ENEMY INVASION

Our New Ally Now in Possession of the Frontier Cost of the Danube—Roumania Invasion of Austria Meeting With Success.

SESSION OF HUNGARIAN DIET WAS A VERY STORMY ONE

Brunswick News Digitization

- 2,500,000 Pages
- Source : Microfilm



TELEGRAPH-JOURNAL

GALLANT URGED TO 'PUSH' PM TO STAND UP FOR PIPELINE

Atlantic Liberal gathering is part regional summit, part fundraiser

Arts board wants to meet with premier over axed funds

The cute little bunnies that decimated an island ecosystem

inside

A primary source for all your vision, eye health and eyewear needs.

I Think We Do Great Things!

(Apologies for speed and brevity)

Digitization?

Table 2. Framework for a Digitization Project

Creating digital collections encompasses a diverse array of activities. The list of main functional areas that follows represents a wide range of skills:

Selection
- material selection based on research, learning, and teaching needs
- copyright-status assessment

Requirements analysis to set technical requirements for:
- digitization
- metadata
- access and use
- other repurposing areas (e.g., print on demand)

Preparation
- conservation, disbinding, tagging
- physical volume organization by content or format type

Digitization
- digitization (in-house or outsourced)
- image processing
- creation of archival and derivative files
- structuring

Quality control
- development of a QC strategy
- selection of QC tools
- development of assessment workflow
- plan for correcting and reintegrating unacceptable images and other deliverables

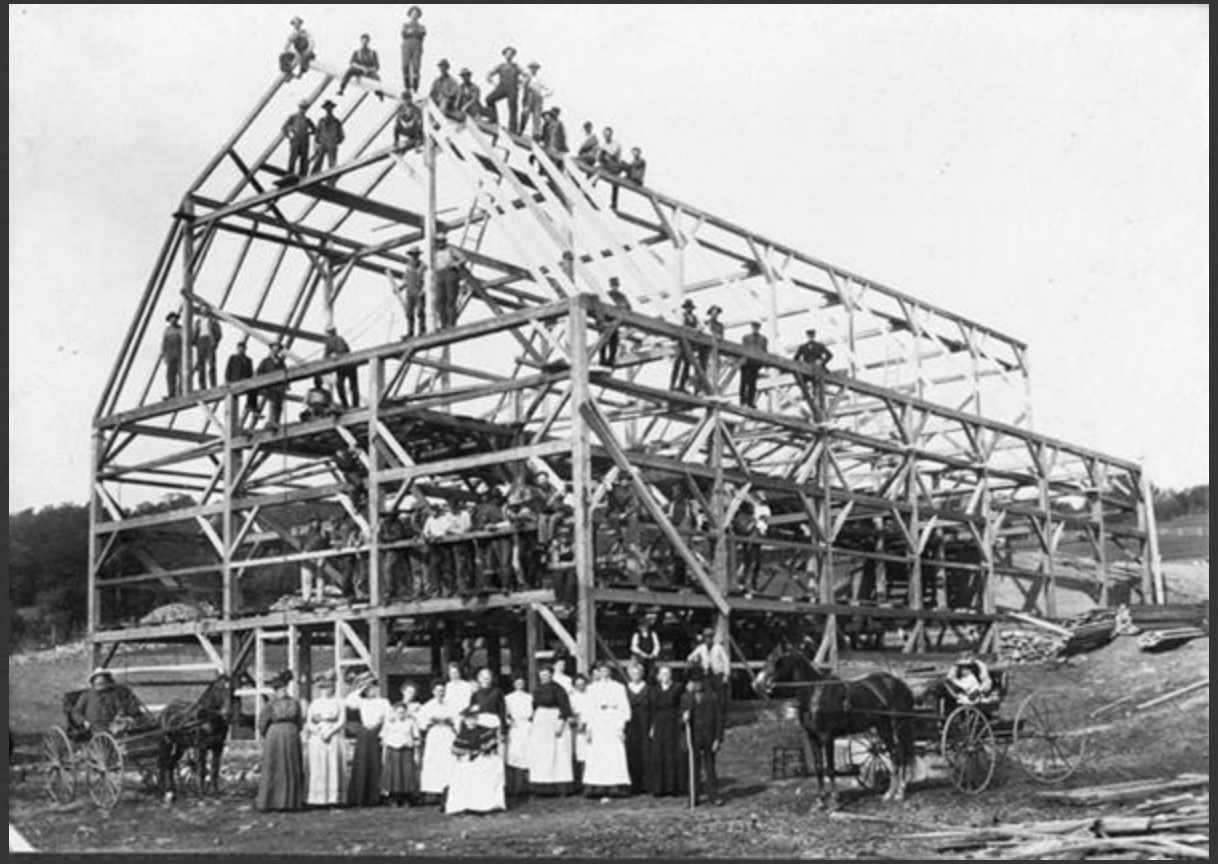
Metadata
- descriptive, structural, administrative, preservation
- controlled vocabulary, taxonomies, ontologies
- selecting and implementing standards for interoperability, discovery, etc.
- file-naming conventions and persistent IDs
- OCR

Technical development
- repository and storage plan
- digital content delivery platform (image database)
- discovery and navigation tools
- Web services
- Web design and development

Project management
- workflow coordination
- financial management
- assessment and usability analysis
- promotion
- user support

Life cycle management
- preservation strategies and procedures
- ongoing content, metadata, application revisions, additions, etc.

Oya Y. Rieger (2008) - Preservation in the Age of Large-Scale Digitization



Alexander W. Galbraith - This image is available from the City of Toronto Archives

Table 2. Framework for a Digitization Project

Creating digital collections encompasses a diverse array of activities. The list of main functional areas that follows represents a wide range of skills:

Metadata

- file-naming conventions and persistent IDs
- OCR

Technical development
- repository and storage plan
- digital content delivery platform (image database)
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Digitization

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Oya Y. Rieger (2008) - Preservation in the Age of Large-Scale Digitization

Content Dissemination

Asset Archive

Derivative Processing

1. Content Dissemination

The Loyalist Collection

FIDDLEHEAD

Electronic Text Centre

Contested Terrain

Atlantic Canada Virtual Archives

DIGITALScholarship journals

Passamaquoddy Maliseet Language Portal
Welcome to the Passamaquoddy-Maliseet Language Portal

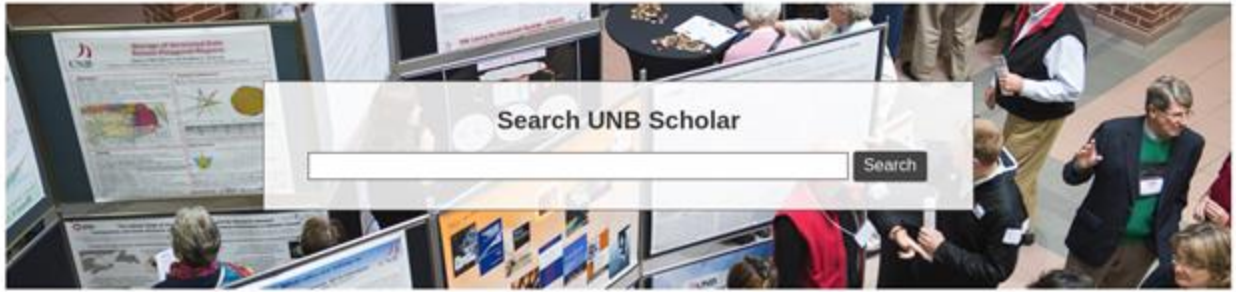
How did we do it?

Early on : mix of tools.

(later):

UNB LIBRARIES | SCHOLAR RESEARCH REPOSITORY

Collections Browse About



About UNB Scholar

UNB Scholar is an institutional repository initiative of UNB Libraries intended to collect, preserve, showcase, and promote the open access scholarly output of the UNB community. Use UNB Scholar to explore specific collections, or search all content in the repository. Material submitted to the repository will also be freely discoverable online through Google and other major search engines.

UNB Libraries welcomes submissions to UNB Scholar. Find out more about depositing your publications!

Browse Collections

Faculty / Staff Works

Theses and Dissertations

Undergraduate Senior Reports

Books

New in UNB Scholar

Passion and Persistence: An Inaugural History of the New Brunswick College of Craft and Design

The dynamics of empowerment measures on population health outcome and well-being

Developing a communication checklist to enhance patient satisfaction with total knee replacement surgery: a mixed methods study

An exploratory study of knowledge-seeking behaviour of New Brunswick entrepreneurs

Career decision-making self-efficacy among higher education women and its associations with type and perceived quality of intimate relationship: exploring the links

Translators for interoperating and porting object-relational knowledge

UNB LIBRARIES | SCHOLAR RESEARCH REPOSITORY

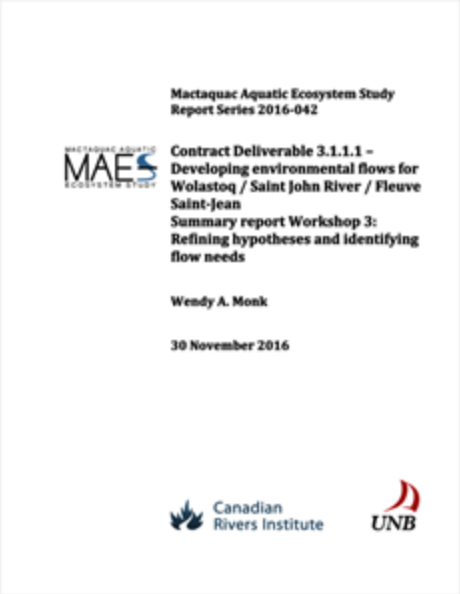
Search

Collections Browse About

Home » Islandora Repository » Faculty / Staff Works » Research Institutes & Centres » Canadian Rivers Institute » Mactaquac Aquatic Ecosystem Study Report Series

Mactaquac Aquatic Ecosystem Study Report Series 2016-042, Contract Deliverable 3.1.1.1 – Developing environmental flows for Wolastoq / Saint John River / Fleuve Saint-Jean, Summary report Workshop 3: Refining hypotheses and identifying flow needs

Monk, W. A. (2016). Mactaquac Aquatic Ecosystem Study Report Series 2016-042, Contract Deliverable 3.1.1.1 – Developing environmental flows for Wolastoq / Saint John River / Fleuve Saint-Jean, Summary report Workshop 3: Refining hypotheses and identifying flow needs.



Mactaquac Aquatic Ecosystem Study Report Series 2016-042

Contract Deliverable 3.1.1.1 – Developing environmental flows for Wolastoq / Saint John River / Fleuve Saint-Jean
Summary report Workshop 3: Refining hypotheses and identifying flow needs

Wendy A. Monk

30 November 2016

Canadian Rivers Institute

UNB

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PDF
Details

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Search

Collections Browse About

Home » Islandora Repository » Faculty / Staff Works » Faculty of Arts (Fredericton) » Department of Economics (Fredericton) » Working Papers Series

Did the 1994/96 Employment Insurance Reforms Improve Labour Market Outcomes for Young People

Audas, R., & Murrell, D. (no date). Did the 1994/96 Employment Insurance Reforms Improve Labour Market Outcomes for Young People.

DID THE 1994/96 EMPLOYMENT INSURANCE REFORMS IMPROVE LABOUR MARKET OUTCOMES FOR YOUNG PEOPLE?

by

Rick Audas
&
David Murrell

Working Paper Series 2001-02

UNB

DEPARTMENT OF ECONOMICS
THE UNIVERSITY OF NEW BRUNSWICK
FREDERICTON, CANADA

Download
PDF

Description

One of the goals of the Employment Insurance (EI) changes during 1994/96 was to reduce the number of habitual, seasonal EI users, and to re-channel such users into higher skilled, lower-unemployment occupations. The changes were expected to re-direct a large proportion of young people into more productive human capital acquisition and occupational-choice activity. This paper investigates, using a simple one-equation model, the factors associated with a polychotomous (multinomial) variable describing labour-market states for young people. The model uses year / province interaction variables to explain pre- and post-reform EI policy changes, along with labour-market and socioeconomic control variables. This paper finds that labour market/human capital participation for young people improved steadily, from the 1980s through the late 1990s, for young people living as dependants within a family. But for young people living away from their parents, there was little long-run economic improvement. For this group, there is some evidence that the 1994/96 EI reforms did play a small role in improving labour market / education outcomes.

Details

We digitized quality, original content.

We had the tools to archive and disseminate it.

Something wasn’t right.

Dissemination Challenges:

- UI Customization
- Maintenance
- Sustainability

Issues with UI / Customization:

Java, Java, Java!

Dropping Feature Requests

Platform Abstraction

Issues w/ Maintenance & Sustainability:

- One VM per project / task
- VMs custom built. Complex.
- Struggling to maintain VMs

Maintenance & Sustainability: Issues

- Derivative processed on web instances
  - Microservices helped!

Renaissance...

Declaring Bankruptcy on Technical Debt: Modernizing a Library Systems Infrastructure and Workflow (2016)

Bankruptcy?

- Drowning in technical debt
- We needed to change!

Direct Actions, Improvements:

- *Front-End* : Contract to core frameworks
- *Storage* : Sanity check
- *Hardware* : Reduce Technical Burden of Maintaining of VMs

Looking Forward : Philosophy

- Focus on expertise
- Get better
- Jettison anchors

Looking Forward : Action

Minimize hardware and VMs

Work with far fewer frameworks and platforms

1. Minimize Hardware

(Obligatory) AWS/cloud is great!

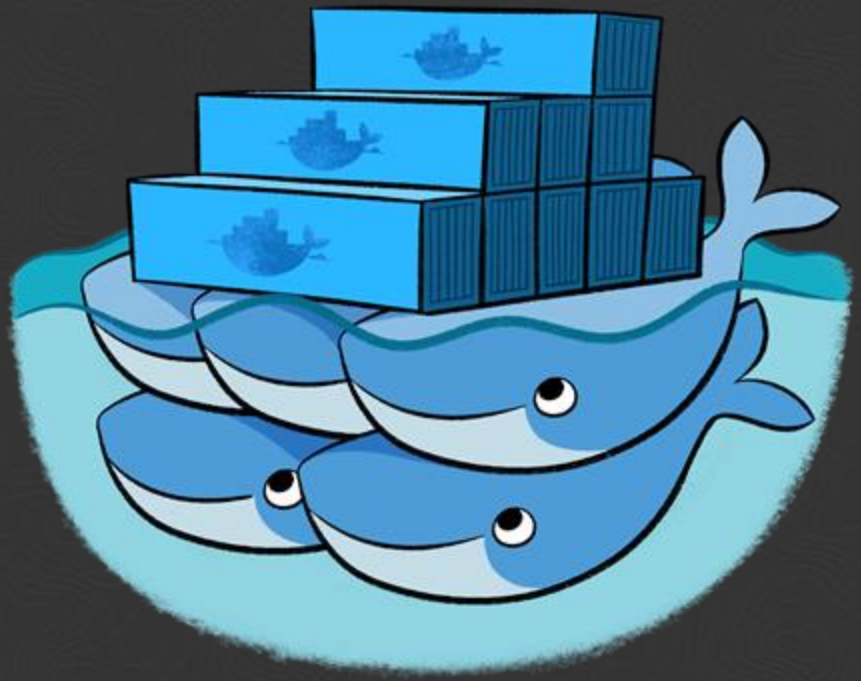
Stop Obsessing about Specific Hardware!

PERCEPTION STOP!

Projects are tenants of SPECIFIC hardware

PERCEPTION START!

Projects are applications that are unlinked to specific hardware.



Container Based Infrastructure

(Virtual Machines)

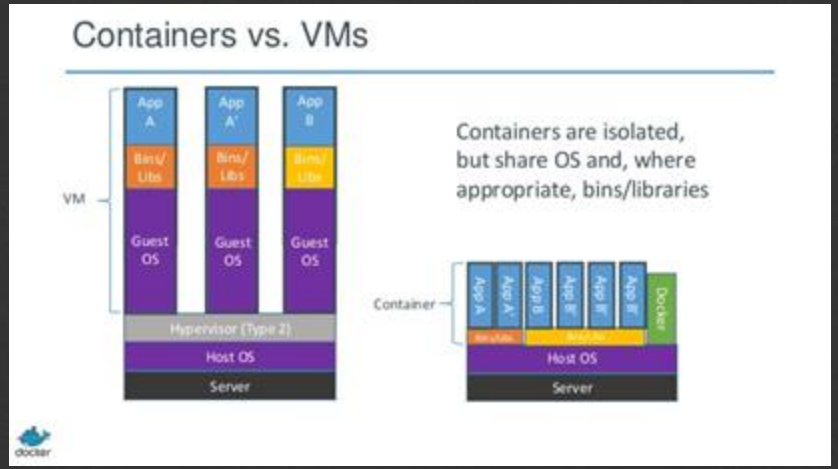
Isolated Instance

High Level Abstraction

(Containers)

Isolated Instance

OS level abstraction



docker.com

Docker:

Runs applications exactly the same each time.

Regardless of the hardware or host.

(Orchestration Issues)

CoreOS

Open Source Projects for Linux Containers

CoreOS Blog

← Back to All Blogs

Container orchestration: Moving from fleet to Kubernetes

February 07, 2017 • By Josh Wood

Tags: fleet Kubernetes

Over the past two years, we've seen a shift in the way organizations think about and manage distributed applications. At CoreOS, work toward this shift began with fleet, a simple distributed service manager released in 2014. Today, the community is seeing widespread adoption of Kubernetes, a system with origins at Google that is becoming the de facto standard for open source container orchestration.

For numerous technical and market reasons, Kubernetes is the best tool for managing and automating container infrastructure at massive scale.

To this end, CoreOS will remove fleet from Container Linux on February 1, 2018, and support for fleet will end at that time. fleet has already been in maintenance mode for some time, receiving only security and bugfix updates, and

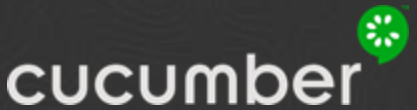
Frameworks

Framework / Language Standardization

(Previously)

Multiple languages, frameworks, significant custom development.

(New)



drupal.org

Why Drupal?

- Our Expertise!
- Rapid data model prototyping
- VCS schema changes
- Fits well in our infrastructure
- Fewer abstractions
- Form API/VCS

DSpace / Fedora Components in Drupal

- UUIDS
  - Entity IDs
- RDF? Triples?
  - RDF Extensions : https://www.drupal.org/project/rdfx
- MODS/XML?
  - Views Data Export : https://www.drupal.org/project/views_data_export
- Disseminators?
  - (Images) Image Styles : https://www.drupal.org/docs/8/core/modules/image/working-with-images
  - (Everything Else) Widgets / Formatters / Entity View Modes https://www.drupal.org/docs/8/api/entity-api/display-modes-view-modes-and-form-modes
- Revisions?
  - Yes

Drupal : Importantly

- Atomic data model
- Future Portability

Search or jump to... / Pull requests Issues Marketplace Explore

unb-libraries / **docker-drupal**

Unwatch 10 Star 2 Fork 2

Code Issues 0 Pull requests 0 Projects 0 Wiki Insights Settings

A lightweight, extensible Drupal 8 PHP-FPM docker image. Suitable for a development-to-production CI oriented workflow. Our base Drupal 8 image at **@unb-libraries**.

Edit

Add topics

365 commits **2** branches **0** releases **1** contributor

Branch: alpine-nginx-p... New pull request Create new file Upload files Find file Clone or download

JacobSanford Allow non-numeric hosts when updating hostname in settings.php Latest commit e38e9bd on Apr 25

build	Set minimum stability to stable across the board	6 months ago
conf	Override php.ini and php-fpm.conf files	2 years ago
env	Remove awkward deprecated container link style of communication	2 years ago
scripts	Allow non-numeric hosts when updating hostname in settings.php	a month ago
tests/behat	Lock symfony to 3.4	6 months ago
travis	Move travis testing upstream and pull from repo	a year ago
.travis.yml	Fix error in triggering Docker Hub build	a year ago
Dockerfile	Remove UserLoginTest from default tests	3 months ago
README.md	Update README.md	8 months ago
docker-compose.yml	Lock local MySQL in compose to 5.7	a month ago

README.md

unblibraries/drupal

81MB 65 layers build passing

A lightweight extensible Drupal docker image, suitable for a development-to-production workflow.

Salix nigra Marsh.

Specimen Data | Edit | Delete | High Resolution Image | Export to CSV | Master Image

Show Full Record

Descriptor	Value
Basis of Record	Preserved Specimen
UNB Accession No.	63201
Collectors	Blaney, C.S. Goltz, J.P.
Other Catalogue No.	6990
Reproductive Condition	No reproductive structures seen on specimen
Previous Identifications	Salix nigra Marsh. by C.S. Blaney
Collection Date	2008-07-27
Event Date	Y: 2008 M: 7 D: 27
Habitat	Sandy river shore under silver maple
Event Remarks	Lower Saint John River Survey Segment #30
Country	Canada
State/Province/Territory	New Brunswick
Locality	Long Island, E side Long Islnad, opposite S end of Lower Musquash Island
Family	Salicaceae
Genus	Salix
Specific Epithet	nigra
Scientific Name Authorship	Marsh.
English Common Name	Black willow
French Common Name	Saule noir
Taxon Remarks	Native

View high resolution image

Download image (JPEG)

View larger map

This website is dedicated to the memory of Harold R. Hinds (1937 - 2001) for his vision and dedication to botany in N.B.

UNB | LIBRARIES

2. Asset Archive

Asset Storage : Before

Fedora / DSpace Native Assetstore

On-VM storage or NFS

No longer using these tools

Asset Storage: Requirements

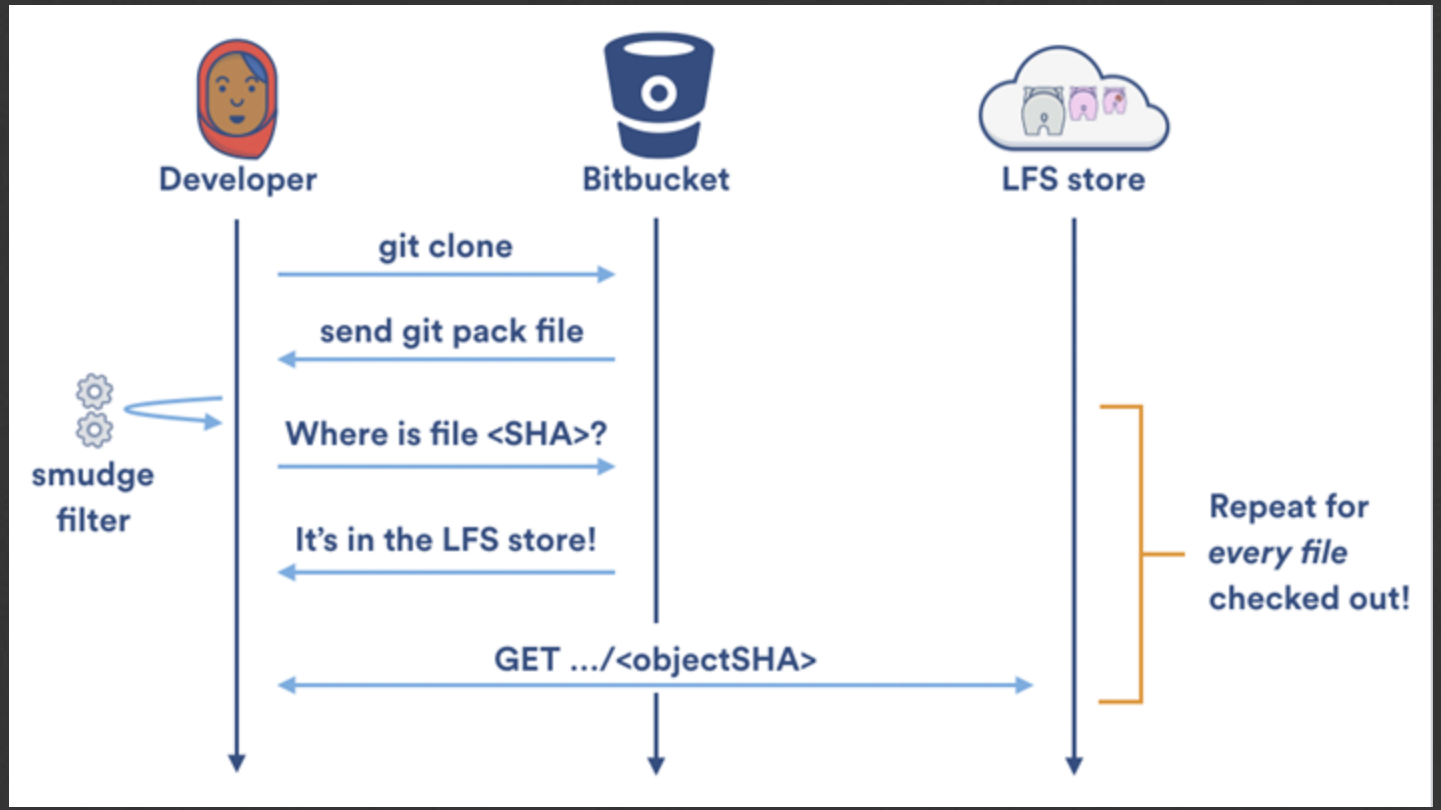
- Decoupled from front-end
- Version Control
- Drupal Integration
- Integrity checking

git-scm.com

Git LFS

“Git Large File Storage (LFS) replaces large files such as audio samples, videos, datasets, and graphics with text pointers inside Git, while storing the file contents on a remote server like GitHub.com or GitHub Enterprise.”

Git LFS



dzone.com

Git LFS

Search or jump to... / Pull requests Issues Marketplace Explore

unb-libraries / herbarium-specimen-masters Private

Unwatch 6 Star 0 Fork 0

Code Issues 0 Pull requests 0 Projects 0 Wiki Insights Settings

Branch: master herbarium-specimen-masters / 10159.tif Find file Copy path

webupdate [10159] Batch import of archival file. 42ae89a on Aug 3, 2017

1 contributor

Executable File | 4 lines (3 sloc) | 190 MB Raw Blame History

```
1 version https://git-lfs.github.com/spec/v1
2 oid sha256:b6788e892746307da2e23f1ed4d1f961cf8d9b2a60b44abf043ed50edb256ce7
3 size 199411516
```

Git LFS

- Version Controlled
- Hash as OID - easy integrity checks
- No new tools - we already use git!

Master Image

Specimen Data | Edit | Delete | High Resolution Image | Export to CSV | Master Image

Information

The specimen master image serves as the digital preservation copy for the herbarium specimen. The images are stored in a separate repository and version controlled to ensure integrity. Additionally, all local images used in the site are derived from this master image.

Master Image Audit History

Revision	Date	User	Change	Commit
1	2017-08-03 23:42:48 +0000	libsystems@unb.ca	[4188] Batch import of archival file.	d696b2715
2	2018-06-01 17:15:44 +0000	libsystems@unb.ca	[4188] Upload of new archival file.	e9d66d4df
3	2018-06-01 17:58:08 +0000	libsystems@unb.ca	[4188] Upload of new archival file.	154f234f6
4	2018-06-01 18:55:23 +0000	libsystems@unb.ca	[4188] Upload of new archival file.	9f03cde5b
5	2018-06-04 16:10:21 +0000	libsystems@unb.ca	[4188] Upload of new archival file.	024739d44
6	2018-06-04 16:19:25 +0000	libsystems@unb.ca	[4188] Upload of new archival file.	4ae01a4f9
7	2018-06-05 19:11:37 +0000	libsystems@unb.ca	[4188] Deletion of archival file.	4f54efbd7

Upload New Master Image

If you wish to add a new or replace an existing master image, upload one below. Adding a new master will trigger several tasks and can take upwards of several minutes.

The master image file should be the original TIF file from the scanner, unmodified.

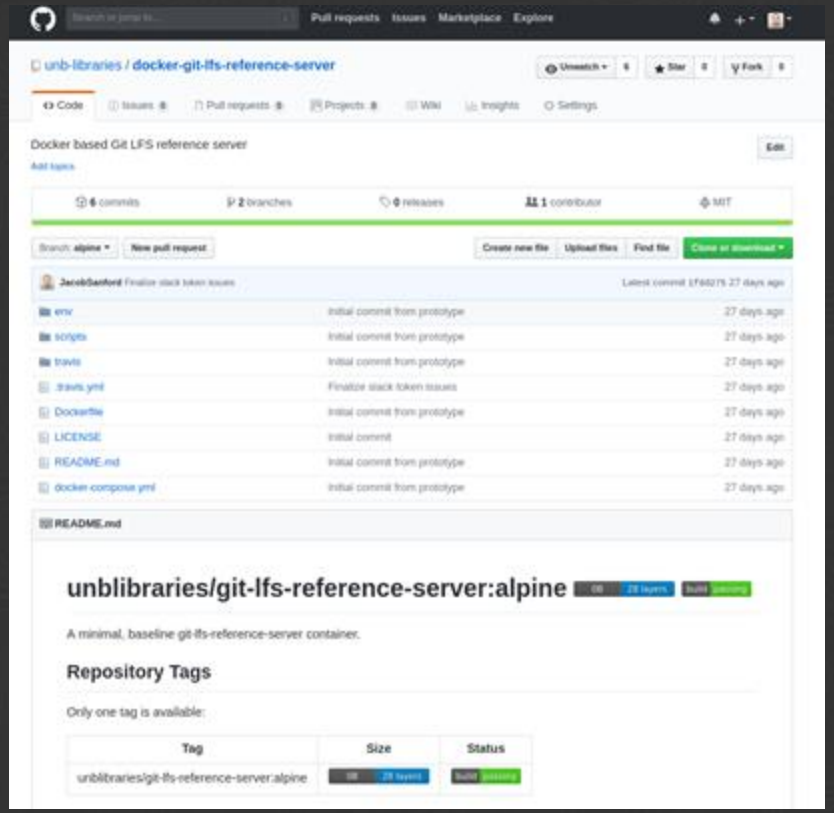
TIF File*

Choose File No file chosen

Upload an master image, allowed extensions: TIF TIFF

Upload New Master Image

Git LFS


Pull requests
Issues
Marketplace
Explore
unb-libraries / docker-git-lfs-reference-server
Code
Docker based Git LFS reference server
Edit
6 commits
2 branches
0 releases
1 contributor
MIT
Branch: alpine
New pull request
Create new file
Upload files
Find file
Clone or download
JacobSanford
Latest commit
env
scripts
travis
.travis.yml
Dockerfile
LICENSE
README.md
docker-compose.yml
Initial commit from prototype
Finalize stack token issues
Initial commit
27 days ago
README.md
unblibraries/git-lfs-reference-server:alpine
A minimal, baseline git-lfs-reference-server container.
Repository Tags
Only one tag is available:
Tag
Size
Status
unblibraries/git-lfs-reference-server:alpine

3. Derivative Processing

Digital Capture

- Asset capture : Center for Digital Scholarship

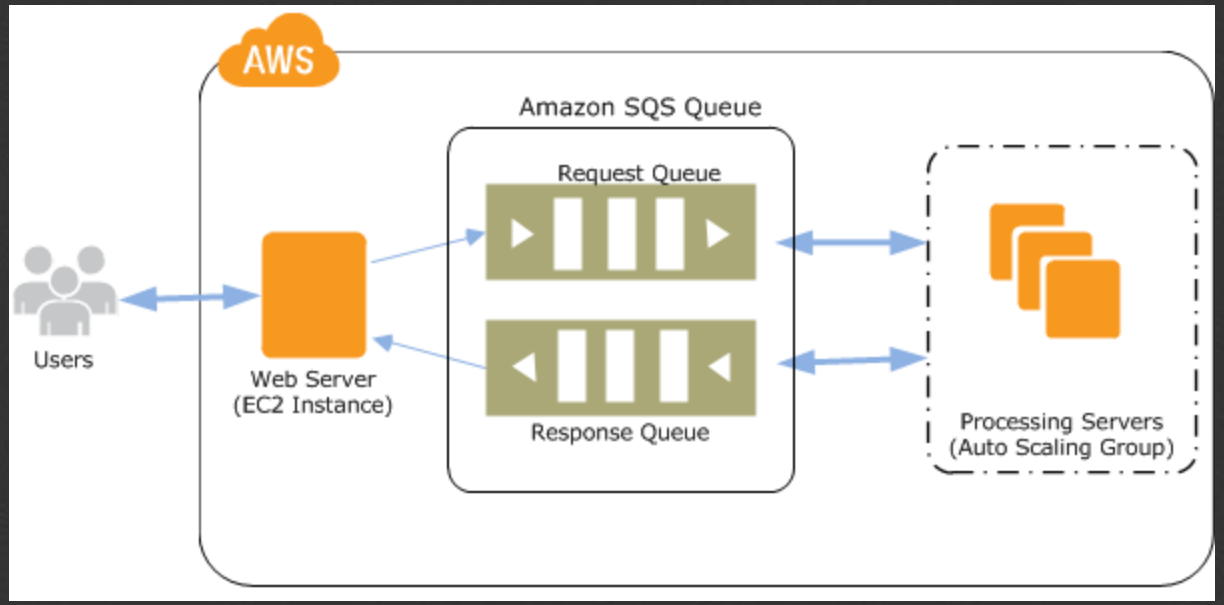
Derivative Processing Workflow:

- Purchase / Allocate Storage
- Purchase / Allocate Processing
- Deploy and Configure Processing Hardware
- Create custom processing workflow
- Parallelize the tools in workflow
- Queue images
- Begin derivative processing
- Store derivatives
- Disseminate

1. **Queue Images**

2. **Process From Queue**

Queue Images



Process Images

unb-libraries / herbarium-generate-jp2-surrogates

Unwatch 5 | Star 0 | Fork 0

Code | Issues 0 | Pull requests 0 | Projects 0 | Wiki | Insights | Settings

Branch: master | herbarium-generate-jp2-surrogates / herbariumGenerateSurrogates.sh | Find file | Copy path

JacobSanford Graphicsmagick changes — a4253a8 on Jun 18, 2014

1 contributor

Executable File 52 lines (42 sloc) 2.22 KB — Raw | Blame | History

```
# Lib Stuffs
export JAVA_HOME=/usr/lib/jvm/java-7-oracle/jre
export LD_LIBRARY_PATH=/opt/kakadu
export CATALINA_HOME=/usr/share/tomcat6
export KAKADU_HOME=/opt/kakadu
export OMP_NUM_THREADS=32

# Path Stuffs
inputdirtouse='/root/toProcess'
outputdirtouse='/root/convertOutput/herbarium-output'
tempdirtouse='/root/convertTemp'
kakadubindir='/opt/kakadu/bin'
convertbindir='/usr/bin'

# Overlay To Use
overlayfiletomask='./31002100.png'

# Set Up Loop
find $inputdirtouse -iname "*.tif" -and -not -iname "*_*.tif" | while read curfilepath
do
        # Define Vars for This Run
        curfilenametouse=`basename $curfilepath`
        curfileroottouse=`echo $curfilenametouse | sed 's/\.tif//g'`
```

Queue Images

unb-libraries / BNIFileProcessing

Unwatch 8 | Star 0 | Fork 0

Code | Issues 0 | Pull requests 0 | Projects 0 | Wiki | Insights | Settings

BNIFileProcessing

Edit

Add topics

30 commits | 1 branch | 0 releases | 1 contributor

Branch: master | New pull request | Create new file | Upload files | Find file | Clone or download

JacobSanford Update README.md — Latest commit c340d2c 13 seconds ago

BNIImage	Remove UUID from filename	2 years ago
.gitignore	Add gitignore	2 years ago
README.md	Update README.md	13 seconds ago
archiveToHiStorage.sh	Add helper scripts to repo	2 years ago
mount_s3fs.sh	Initial commit from prototype	3 years ago
processBNIFiles.sh	Add helper scripts to repo	2 years ago
processBNIImages.py	Further changes	2 years ago
umount_s3fs.sh	Initial commit from prototype	3 years ago

README.md

BNIFileProcessing

General Info

Serves as the core tool for processing newspaper images scanned in our large digitization project. Most likely not useful for you.

Derivative Processing : Challenges

- Portability
- Scalability
- Dependencies
- Re-Using Tools

Processing: Hardware Obsession

PERCEPTION STOP!

Digital asset processing as tenants of servers

PERCEPTION START!

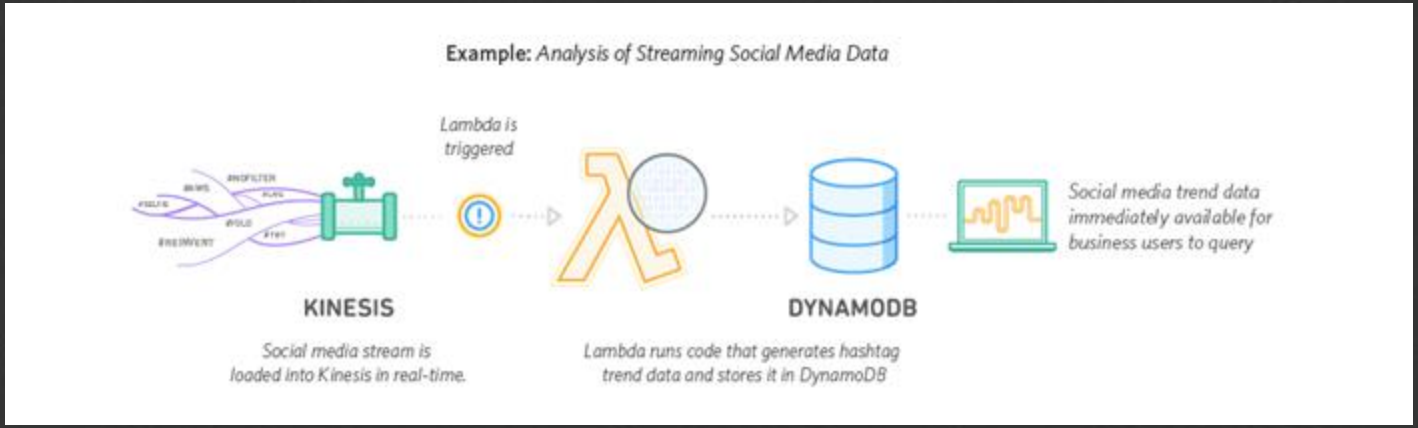
Digital asset processing is simply series of discrete **serverless** functions.

They will run somewhere. It doesn’t matter. Stop caring!

Serverless Functions

FAAS

AWS Lambda

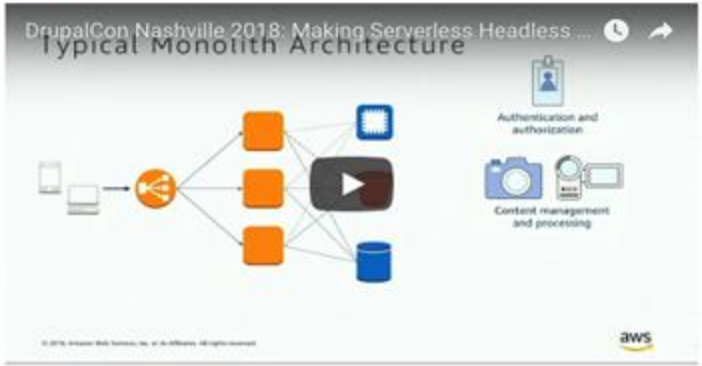

Example: Analysis of Streaming Social Media Data
Lambda is triggered
KINESIS
Social media stream is loaded into Kinesis in real-time.
DYNAMODB
Lambda runs code that generates hashtag trend data and stores it in DynamoDB
Social media trend data immediately available for business users to query

DIAMOND SPONSORS

platform.sh Acquia PANTHEON

Making Serverless Headless - Drupal with AWS Lambda@Edge & Amazon CloudFront

restrictedinfinity
naligoel



AWS Lambda@Edge allows you to run code closest to your end customers. In addition to being a serverless platform, it globally distributes your web-application to run out of dozens of AWS data centers across the planet, with your customers being served from the one nearest to them. Come join us for this session to learn how you can combine its forces -- with Drupal as powerful Headless CMS, Lambda@Edge running Javascript frontend framework (like Angular, Ember and React), and CloudFront as Content

Session Track
Ambitious Digital Experiences

Experience level
Beginner

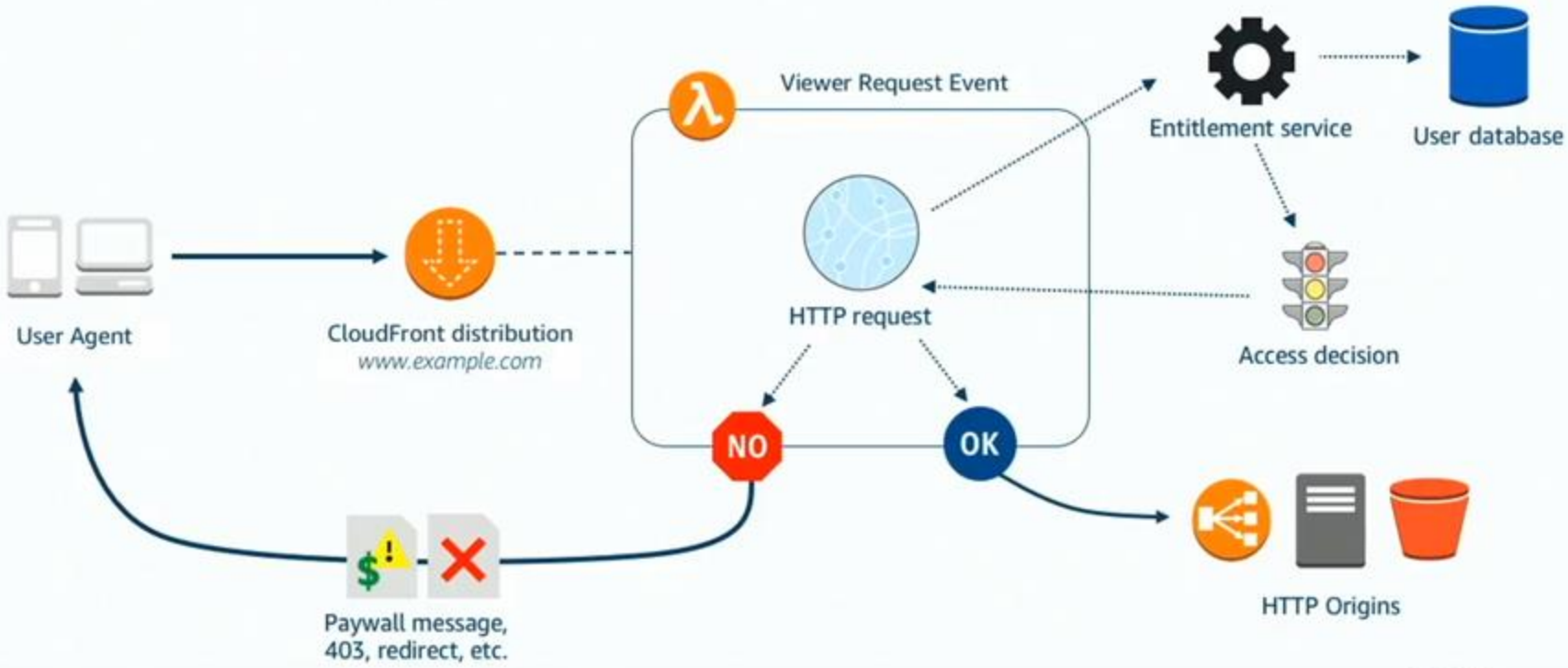
Drupal Version
Drupal 8 (current)

Session Time Slot(s)
Time:
04/11/2018 - 17:00 to 04/11/2018 - 18:00
Room:
207AB | Phase2

Remove from your schedule

https://events.drupal.org/nashville2018/sessions/making-serverless-headless-drupal-aws-lambdaedge

VIEWER REQUEST: AUTHORIZATION



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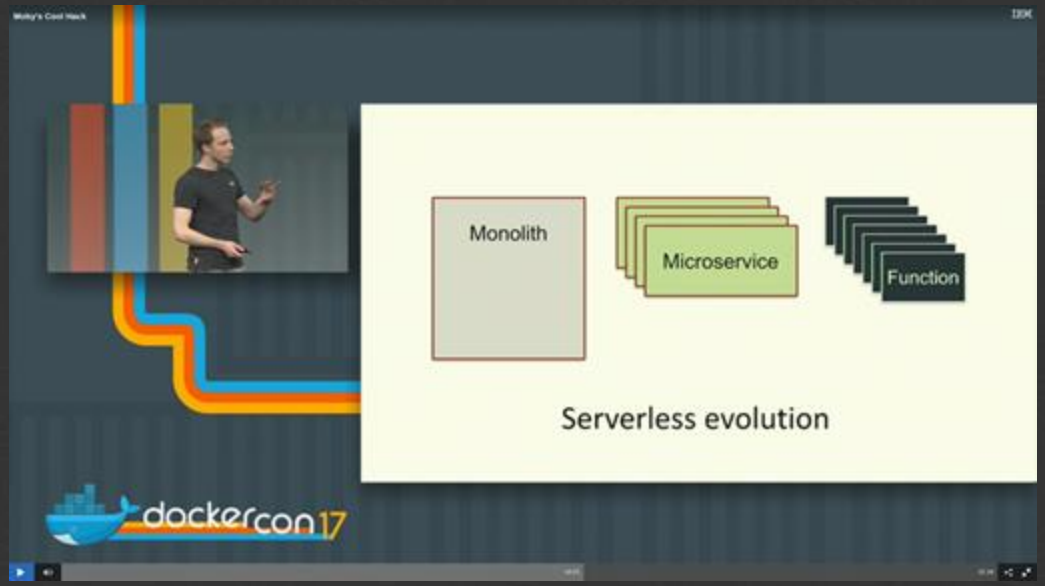
https://events.drupal.org/nashville2018/sessions/making-serverless-headless-drupal-aws-lambdaedge

AWS Lambda

- It's Great!
- Not ideal for our large scale digitization projects.

OpenFAAS

- Alex Ellis : Closing Keynote at DockerCon



https://blog.docker.com/2017/04/dockercon-2017-mobys-cool-hack-sessions/

OPENFAAS

Serverless Functions Made Simple for Docker and Kubernetes

VIEW ON GITHUB

Video Demo

Functions as a Service

API Gateway

Function Watchdog

Prometheus

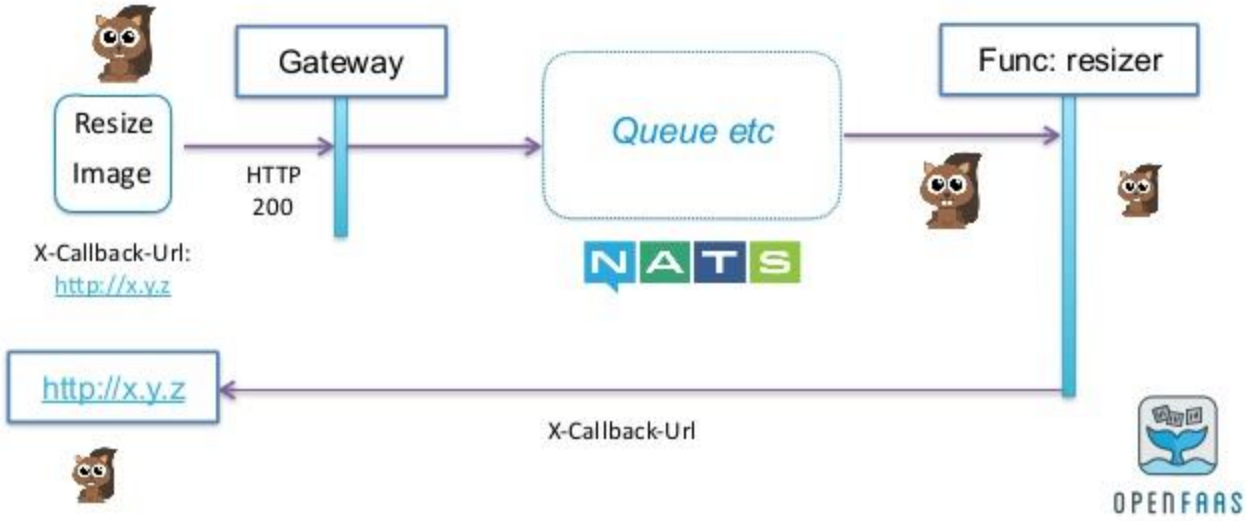
Swarm

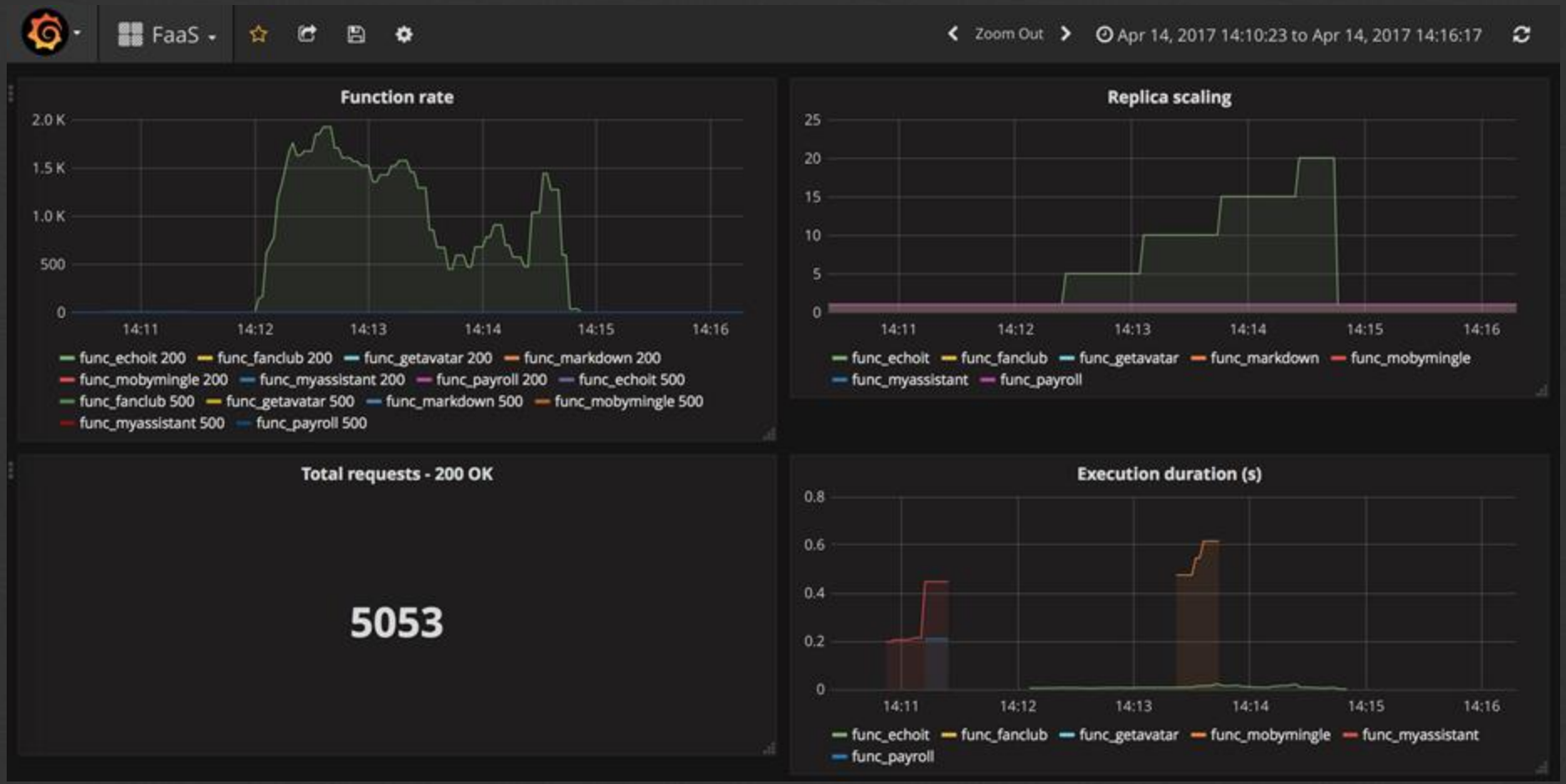
Kubernetes

docker

Asynchronous invocation

DEFERRED EXECUTION




FaaS
Zoom Out
Apr 14, 2017 14:10:23 to Apr 14, 2017 14:16:17
Function rate
2.0 K
1.5 K
1.0 K
500
0
14:11
14:12
14:13
14:14
14:15
14:16
func_echoit 200
func_fanclub 200
func_getavatar 200
func_markdown 200
func_mobymingle 200
func_myassistant 200
func_payroll 200
func_echoit 500
func_fanclub 500
func_getavatar 500
func_markdown 500
func_mobymingle 500
func_myassistant 500
func_payroll 500
Replica scaling
25
20
15
10
5
0
14:11
14:12
14:13
14:14
14:15
14:16
func_echoit
func_fanclub
func_getavatar
func_markdown
func_mobymingle
func_myassistant
func_payroll
Total requests - 200 OK
5053
Execution duration (s)
0.8
0.6
0.4
0.2
0
14:11
14:12
14:13
14:14
14:15
14:16
func_echoit
func_fanclub
func_getavatar
func_mobymingle
func_myassistant
func_payroll

openfaas / faas

Watch 404 | Star 10,304 | Fork 726

Code | Issues 88 | Pull requests 7 | Insights

Branch: master | faas / sample-functions /

Create new file | Upload files | Find file | History

lyovcheva and alexellis Update alpine version to 3.7 ... Latest commit 7f6295c on Apr 30

..

AlpineFunction	Update alpine version to 3.7	24 days ago
ApiKeyProtected-Secrets	Update alpine version to 3.7	24 days ago
BaseFunctions	Update watchdog version to 0.8.0	24 days ago
CHelloWorld	Update alpine version to 3.7	24 days ago
CaptainsIntent	Update alpine version to 3.7	24 days ago
ChangeColorIntent	Update alpine version to 3.7	24 days ago
DockerHubStats	Update alpine version to 3.7	24 days ago
HostnameIntent	Update alpine version to 3.7	24 days ago
MarkdownRender	Update alpine version to 3.7	24 days ago
Nmap	Add nmap sample	5 months ago
NodeHelloEnv	Remove un-needed shell files	5 months ago
NodeInfo	Update alpine version to 3.7	24 days ago
Phantomjs	Update watchdog version to 0.8.0	24 days ago
ResizeImageMagick	Update alpine version to 3.7	24 days ago
SentimentAnalysis	Update watchdog version to 0.8.0	24 days ago
WebhookStash	Update alpine version to 3.7	24 days ago
WordCountFunction	Config tweak for samples	3 months ago
apikey-secret	Fix apikey-secret README.md	2 months ago
echo	rename catservice	8 months ago
figlet	Update alpine version to 3.7	24 days ago
gif-maker	Update alpine version to 3.7	24 days ago
pwgen	Add pwgen and bump Alpine base image	2 months ago

viveksyngh / openfaas-ocr

Watch 1 | Star 1 | Fork 1

Code | Issues 0 | Pull requests 0 | Projects 0 | Wiki | Insights

Tesseract OCR function for OpenFass

12 commits | 1 branch | 0 releases | 1 contributor

Branch: master | New pull request | Create new file | Upload files | Find file | Clone or download

viveksyngh Merge branch 'master' of https://github.com/viveksyngh/openfaas-ocr — Latest commit a071308 on Jan 25

logo	added logo	4 months ago
openfaas-ocr	applied check for missing url type	4 months ago
screens	added screens	4 months ago
.gitignore	add file download from url and file type check	4 months ago
README.md	Update README.md	4 months ago
openfaas-ocr.yml	update gateway url	4 months ago

README.md

openfaas-ocr

Tesseract OCR function for OpenFass

Tesseract is an optical character recognition engine for various operating systems. It is free software, released under the Apache License, Version 2.0, and development has been sponsored by Google since 2006. In 2006 Tesseract was considered one of the most accurate open-source OCR engines then available.

Note: Input for this is an URL of image with .jpg, .jpeg or .png extension.

Usage

Deploy:

```
$ faas-cli deploy -f ./openfaas-ocr.yml --gateway=http://<GATEWAY-IP>
```

Sam Davern
@DavernSam

@colorisebot

8:33 PM - Nov 4, 2017

See Sam Davern's other Tweets

Colorise Bot
@colorisebot

Replying to @DavernSam

I colorized your image using #openfaas in 4.9 seconds #dockercon

8:33 PM - Nov 4, 2017

3 See Colorise Bot's other Tweets

Derivative Processing with OpenFAAS

- Offload heavy lifting to FAAS cluster
  - Initial bulk ingest
  - New items through interface
    - Drupal modules : Guzzle

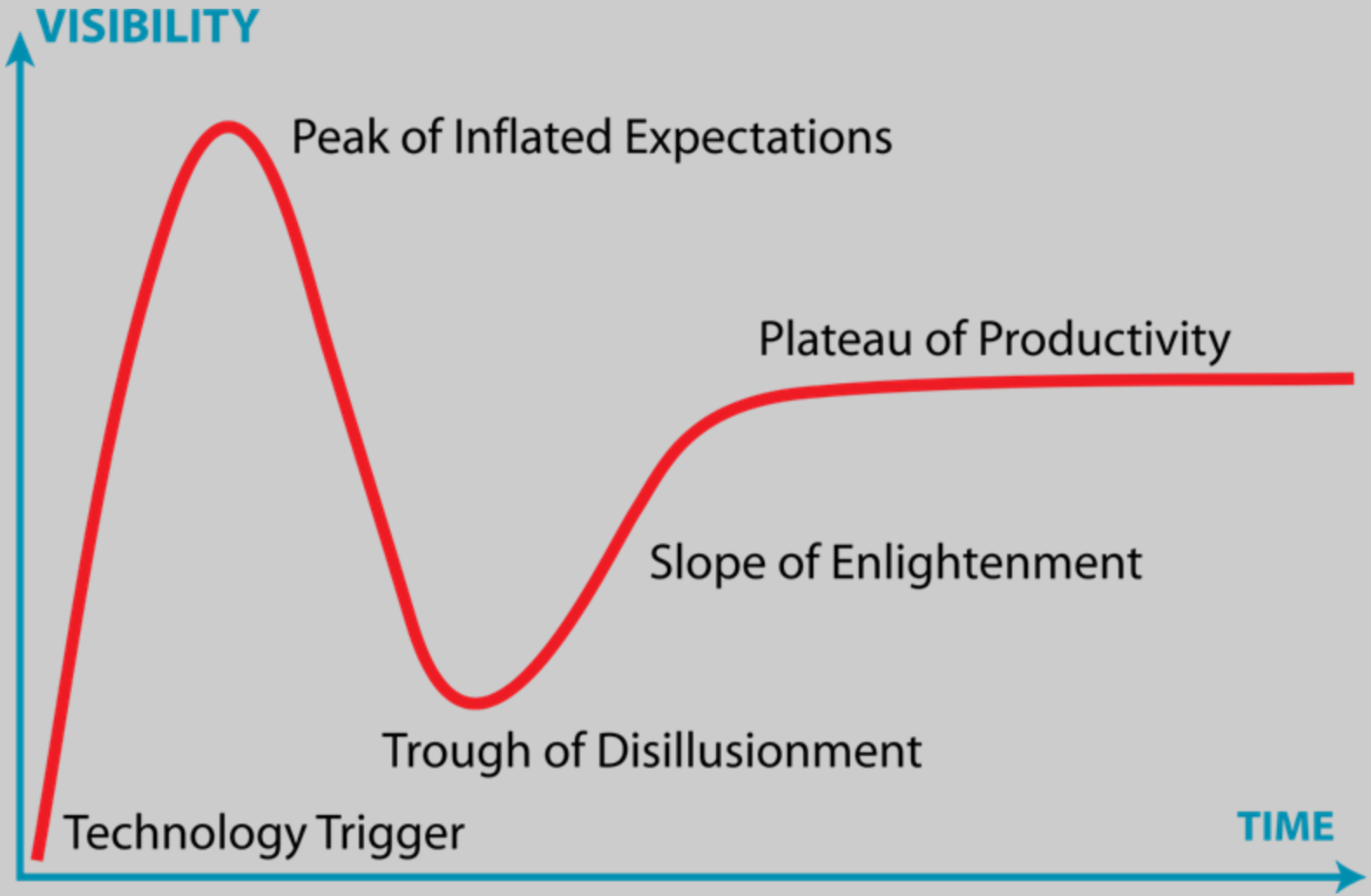
Derivative Processing with OpenFAAS

- Developer experience
  - Queue and process derivatives from a workstation

Derivative Processing with OpenFAAS

- This is not a novel concept!
  - Microservices/messaging well-established
    - OpenFAAS fits like a glove into our infrastructure and philosophy.

Status?


VISIBILITY
Peak of Inflated Expectations
Plateau of Productivity
Slope of Enlightenment
Trough of Disillusionment
Technology Trigger
TIME

FAAS Functions : Next Steps

- Establish our standardized library of abstracted functions
- Integrate well with Drupal modules and callbacks
- Determine preferred method for defining meta-functions.

Content Dissemination

Asset Archive

Derivative Processing

Discussion

Acknowledgements

Jeff Carter - Manager, Systems Group

James MacKenzie - Director of Scholarly Technologies

Mario Tiozzo - Director, Finance & Facilities

Lesley Balcom - Dean of Libraries

Steve Rosenfeld - Director of Library Systems (Retired)

Discussion